

MyCal

version: 0.02

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Abstract

MyCal is a program to calculate calories from a database of ingredients. It is based on **psql** and **pyside6**

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1 Introduction

In the following I will try to describe the setup of the program, so that in a few months I still can remember what I did.

2 Layout of the program

Currently, the main class is `ControlDB`, which has the methods described below. While in the beginning I created the form for ingredients from scratch, I later started to use the QT designer. The advantage of the QT designer is that I can easily add or move things in a design, run the conversion from ui to py and be done. Calling the designer is done by **`pyside6-designer Recipe.ui`** & to open the `Recipe.ui` form. After having changed the design and saved it, running **`pyside6-uic Recipe.ui -o Recipe.py`** converts the design into a python class, which can be reused.

3 class `NewWindow`

Creates a simple mainwindow, which can then be used to add widgets.

4 class `RecipeModel`

This is created from `QAbstractListModel` and creates the model for the recipe form. This is part of the model view architecture of pyside6 and is used by the method **`MyRecipes.new`**.

5 class `ControlDB`

This is the main class of the program. It is called by

```
class ControlDB(QMainWindow):
```

```

def __init__(self,Title=None,db_name=None,db_user=None,db_system=None,db_pwd =
    super().__init__()
    self.db_name = db_name
    self.db_user = db_user
    self.db_system = db_system
    if db_pwd == None:

```

where the database parameters are passed.

5.1 ConnectDataBase

This establishes the connection to the psql database, using a text file for the password. If the connection fails , the program exits

5.2 SetupLogger

This creates the logger system. It deals with warnings and errors.

5.3 ShowTables

Lists all the tables in the database

5.4 ViewTable

This method sets up the connection to a specific table and you can view this table. You can choose the list of columns which you want to suppress in display.

```

def ViewTable(self,table,suppress_columns=[],editmode=True,
columnwidth=[],Title = None, showTable = True):

```

Calling this method with showTable = False will suppress the display. This is one of the main methods to deal with tables and selecting them.

5.5 SetPosition

Move the window to position chosen.

5.6 CreateIngredientsForm

This creates the form to add ingredients. This builds the respective form from scratch, something I don't want to do . It connects to SaveIngredients and also to Cancel

5.7 InsertRecord

First it creates a tableview with the above ViewTable for the ingredients table and then creates a QSqlQuerymodel. It inserts self.record, which has been created by SaveIngredients, which in turn is a connected method called by clicking on the SaveButton from the ingredient form.

5.8 MyRecipes_new

This is the heart of the communication with the recipes table. This establishes the recipe window creates the RecipeView, as well as the RecipeModel. It calls the GetRecipeList, which creates a list of all the recipe names.

5.9 getSelectedText, getSelectedRecipes

This is a function called through a connector in the Recipe Combobox. First it clears the record, and then fills the record with the table row chosen through the combo box. It uses an emit signal to refresh the list.

5.10 FillRecipeIngredientList

This is currently fairly long for historical reasons. The columns , which contains the ingredients of a specific recipe from the original Libreoffice days looks like this:

177 g Flour unbleached *u000a79 g Butter *u000a33 g Almond flour *u000a480 g Apric

There are a few problems with this:

1. We have spaces in the name of some of the ingredients. I manually changed the ingredients table to have always underscores, but the recipe table is too long.
2. We have characters in which have to do with textformatting (*u000a)

3. It is one string

In this routine we strip out the UTF characters, remove the *, and then create a list in self.recipe_ingredients

Note in vs03 more handling create 2 dimensional list because now we have tableview

5.11 DisplayRecipeList

Adds items to the Recipelists.

5.12 split_and_keep

This is used by FillRecipeIngredientList to separate the long string into a list of values, where the separation character is **g**.

5.13 StripUnicode

removes the u000a from the record.

5.14 GetRecipeList

Gets a list of all the recipes in the recipe table.

5.15 SizeWindow

Creates dimensions of the window selected. it gives the upper corner coordinate and the x and y size.

5.16 SaveIngredients

Connected to from the CreateIngredientsForm, it checks that the new ingredient is unique and then saves the new created ingredient in the table.

5.17 on_item_click

gets connected to QlistView, and called when you click on one of the items.

5.18 save_recipe

Called from save button in recipe form, it saves the current record for ingredients. It then calls pack_record, so that we can update the record.

5.19 pack_record

The record from save_recipe needs to be packed back into a single string.

5.20 update_recipe

called from save recipe it updates the ingredient record.

5.21 update_record

This updates row b with name in table

5.22 do_sql

Executes a sql statement whihc has been formed by the calling method.

6 Some Useful commands

1. `pyide6-designer Recipe.ui &`
2. `pyide6-uic Recipe.ui -o Recipe.py`

7 Notes

There are a few issues I still need to address:

1. Make recipe list editable
2. Populate the textfield in the form like calgram etc
3. Create the calorie calculator
4. Make sure entries in table get checked for upper and lower characters

5. Clean up the code with consistent name scheme.

It is important to only use underscores in names or ingredients. Do not use spaces. All entries are in **gram**. I find the US system of oz, cups etc totally stone age; so will never use that system.

8 Current Status

I think with QListView for the recipes, I boxed myself into a corner. This is the end of vs 0.2 I create now vs 0.3 where the interface is a tableview. Tableview is working, next step is to be able to edit the fields and then save them.

9 Current issues

- The unpacking of a ingredienst list with the new stored value only gives the first entry. Needs to be solved first. fixed
- updating the ingredients in recipe does NOT update any calculated values
- finish new recipe setup
- menu